this prerequisite is not met. Because AH representations are geometrically diverse, it is difficult for models to capture a consistent feature pattern among AH samples, thereby hindering the generalization of refusal responses to unseen AH samples. In contrast, UH representations tend to form distinct cluster, presenting consistent feature patterns that should facilitate easier generalization of the refusal capability.

Experimental Setups To verify the hypothesis, we conduct refusal tuning on LLMs under two settings: 1) **UH Only**, where 1,000 UH samples are paired with 10 refusal templates, and 1,000 FA samples are preserved with their original answers. 2) **AH Only**, where 1,000 AH samples are paired with refusal templates, with 1,000 FA samples again remain unchanged. We then evaluate both models on 200 samples each of FAs, UHs, and AHs. A response matching any refusal template is counted as a refusal, and we report the **Refusal Ratio** as the proportion of samples eliciting refusals. This measures not only whether the model refuses appropriately on UHs and AHs, but also whether it "over-refuses" on FA samples.

Experimental Results Figure 10 shows that training with UHs leads to strong generalization across UHs, with refusal ratios of 82% for LLaMA. However, this effect does not transfer to AHs, where refusal ratios fall to 28%, respectively. Moreover, some FA cases are mistakenly refused (29.5%). These results confirm that UHs share a common activation subspace, supporting generalization within the category, while AHs and FAs lie outside this space. By contrast, training with AHs produces poor generalization. On AH test samples, refusal ratio is only 33%, validating that their subject-specific hidden states prevent consistent refusal learning. Generalization to UHs is also weak (23.5%), again reflecting the divergence between AH and UH activation spaces.

Overall, these findings indicate that **the generalizability of refusal tuning is constrained by the heterogeneous nature of hallucinations**. While UH representations are internally consistent enough to support learnable refusal generalization, AH representations are too geometrically diverse for the model to learn a consistent and robust refusal pattern.

7 Conclusions and Future Work

In this work, we revisit the widely accepted claim that hallucinations can be detected from a model's internal states. Our mechanistic analysis reveals that **hidden states encode whether models rely on their parametric knowledge rather than truthfulness**. As a result, detection methods succeed most reliably when outputs are detached from the input but struggle when hallucinations arise from the same knowledge-recall process as correct answers.

These findings lead to three key implications. First, future evaluations should **report detection performance separately** for Associated Hallucinations (AHs) and Unassociated Hallucinations (UHs), as they stem from fundamentally different internal processes and require distinct detection strategies. Second, relying solely on hidden states is insufficient for reliable hallucination detection. Future research should integrate LLMs with **external feedback mechanisms**, such as fact-checking modules or retrieval-based verifiers, to assess factuality more robustly. Third, future studies should **prioritize improving AH detection**. Because AHs occur more frequently in widely known or highly popular topics (§4.2), their undetected errors pose greater risks to user trust and the practical reliability of LLMs.

Limitations

We identify several limitations of our work.

Focus on Factual Knowledge While our analysis identifies challenging cases of hallucination detection methods, our study is primarily limited to factual completion prompts. It does not extend to long-form or open-ended text generation tasks (Wei et al., 2024; Min et al., 2023; Huang and Chen, 2024). Future work should broaden this investigation to these tasks in order to draw more comprehensive conclusions.

Lack of Analysis on Prompt-based Hallucination Detection Approaches Our analysis focuses on white-box hallucination detection methods based on internal states and two black-box approaches based on external features. We do not include verbalization-based strategies (Lin et al., 2022a; Tian et al., 2023; Xiong et al., 2024; Yang et al., 2024b; Ni et al., 2024; Zhao et al., 2024), such as prompting the model to report or justify

its confidence explicitly, which constitute a different line of approach. Exploring such approaches may offer complementary insights into how models internally represent and express uncertainty.

Applicability to Black-box LLMs or Large Reasoning Models Our study is limited to open-source LLMs. Conducting mechanistic analyses on commercial black-box LLMs is not permitted due to access restrictions. Future work could explore alternative evaluation protocols or collaboration frameworks that enable partial interpretability analyses on such systems. In addition, recent studies (Mei et al., 2025; Zhang et al., 2025) have begun examining the internal states of large reasoning models for hallucination detection, suggesting a promising direction for extending our methodology to models with multi-step reasoning capabilities.

Ethical Considerations

This work analyzes the internal mechanisms of large language models using data constructed from Wikidata (Vrandecic and Krötzsch, 2014), which is released under the Creative Commons CC0 1.0 Universal license, allowing unrestricted use and redistribution of its data. All data are derived from publicly available resources, and no private or sensitive information about individuals is included. We used LLM tools for language polishing.

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